The Method of Moments for VHF/UHF Wire Antennas: From the Pocklington

Electric-Field Integral Equation to a Dual-Band 144–440 MHz Design

An essay for readers working in mathematical physics, functional and numerical analysis, and classical electromagnetism.

Abstract

We give a self-contained derivation of the thin-wire electric-field integral equation (EFIE) for a perfect-electric-conductor (PEC) wire antenna, discretise it by the method of moments (MoM) with piecewise-constant basis functions and point-matching (collocation) testing, and analyse the resulting linear system from a numerical-analysis standpoint. The central object is the Pocklington kernel, in which the scalar- and vector-potential contributions are combined *before* evaluation so that the near-field $1/R^2$ and $1/R^3$ singularities cancel analytically; we show that this kernel is the unique formulation that is both pointwise valid and free of separate endpoint (charge) terms. We then confront the discretisation with the classical limits — the short-dipole radiation resistance, the half-wave dipole impedance, the resonant length, the far-field directivity, and the global power balance — and report what the solver reproduces to within a few percent and where the piecewise-constant basis degrades (very short and electrically long dipoles). As an application we design and simulate a dual-band VHF/UHF open-sleeve dipole for the 144–148 MHz (2 m) and 430–440 MHz (70 cm) amateur bands, computing feedpoint impedance, SWR, radiation pattern, directivity, gain, beamwidth, and front-to-back ratio on both bands. The design is a close-coupled open-sleeve dipole: a centre-fed 2 m driven element with a symmetric pair of 70 cm sleeves in the reactive near-field (± 25 mm $\approx 0.036\lambda$). The solver predicts a well-matched 2 m band (SWR 1.7–2.3) *and* a well-matched 70 cm band (SWR 1.3–1.6): the sleeves act as an impedance transformer, cancelling the driven element's $-j130\ \Omega$ capacitive reactance at 1.36λ through a large mutual coupling ($|Z_{12}| \approx 70\ \Omega$). We contrast this with the far-field (Yagi-like) spacing, where the coupling is negligible (a few ohms, $|Z_{12}| \approx 3\ \Omega$) and the 70 cm band is left mismatched (SWR ≈ 7.5) — illustrating that the *coupling regime*, not the mere presence of a second element, determines the dual-band match.

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1. Introduction

The VHF (30–300 MHz) and UHF (300 MHz–3 GHz) portions of the radio spectrum are the working bands of a large fraction of terrestrial wireless communication, and of the amateur-radio 2 m (144–148 MHz) and 70 cm (430–440 MHz) allocations. At these frequencies the electrical size of a practical wire antenna is of order a wavelength: a half-wave dipole on 2 m is about 1 m long, on 70 cm about 35 cm. This is the regime in which the *thin-wire integral-equation* description is both natural and accurate — the wire radius a is a few millimetres, three orders of magnitude below the wavelength, yet the wire is long enough that the current distribution is not trivial.

Analytical antenna theory handles the short dipole ($L \ll \lambda$) and, with more effort, the half-wave dipole ($L \approx \lambda/2$) in closed form. But a dual-band design — one structure that must behave acceptably at two frequencies a factor of three apart — has no closed-form answer: the current on each element is set by the coupled integral equations on *all* elements, and the feedpoint impedance on the upper band is controlled by the lower-band element seen at an electrical length of 1.36λ. This is a problem whose natural formulation is a (singular)

integral equation, and whose natural solution is a numerical one. The method of moments is the standard discretisation.

The purpose of this essay is twofold. First, to derive the thin-wire EFIE and its MoM discretisation with the rigour that a functional/numerical-analysis reader expects: what the integral operator is, what the basis and testing discretisation do to it, where the singularities live, what the quadrature must resolve, and what the convergence and conditioning look like. Second, to use the resulting solver as a *measuring instrument* on a real design problem, and to report — with the numbers the code actually produces — what is robust, what is a basis artefact, and what is physics.

We work throughout with the engineering time convention $e^{+j\omega t}$ and the free-space Green's function $G(R) = e^{-jkR}/(4\pi R)$, $k = \omega/c$. All numerical results in §5 are produced by the accompanying NumPy-only solver `antenna_mom.py` (Appendix A); the validation battery of §4 is `validate_fixed.py`.

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2. The thin-wire electric-field integral equation

2.1 From the potentials to the tangential field

Let the source be a z-directed surface current $I(z')$ on a straight PEC filament of radius a at transverse position (x_0, y_0), with time dependence $e^{+j\omega t}$. In the Lorenz gauge, $\nabla\cdot\mathbf{A} = -j\omega\mu_0\varepsilon_0\,\Phi$, the fields are expressed through the vector potential $\mathbf{A}$ and scalar potential Φ:

$$\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi, \qquad \mathbf{A}(\mathbf{r}) = \mu_0 \int \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, d^3r', \qquad G(R) = \frac{e^{-jkR}}{4\pi R}.$$

For a z-directed filament the only non-zero potential component is A_z, and $\nabla\cdot\mathbf{A} = \partial A_z/\partial z$, so

$$A_z(z) = \mu_0 \int I(z')\, G(z,z')\, dz', \qquad \Phi = -\frac{1}{j\omega\mu_0\varepsilon_0}\frac{\partial A_z}{\partial z}.$$

The tangential (z) scattered field on the wire is therefore

$$E_z^s(z) = -j\omega A_z(z) - \frac{\partial\Phi}{\partial z} = -j\omega\left[A_z(z) + \frac{1}{k^2}\frac{\partial^2 A_z}{\partial z^2}\right],$$

where we used $k^2 = \omega^2\mu_0\varepsilon_0$. Substituting the potential integral,

$$\boxed{E_z^s(z) = -j\omega\mu_0 \int I(z')\left[G(z,z') + \frac{1}{k^2}\frac{\partial^2 G(z,z')}{\partial z^2}\right] dz'} \tag{1}$$

with z the *observation* coordinate. The perfect-conductor boundary condition, $E_z^i(z) + E_z^s(z) = 0$ on the wire surface, gives the EFIE

$$j\omega\mu_0 \int I(z')\left[G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2}\right] dz' = E_z^i(z). \tag{2}$$

Equation (2) is the starting point. It is a Fredholm integral equation of the first kind for the unknown current $I(z')$, with a kernel that is *weakly singular but not integrable in the naive form*: the second-derivative term $\partial^2 G/\partial z^2$ carries $1/R^3$, $1/R^4$, $1/R^5$ pieces that blow up as the observation point approaches the source ($R \to \rho$, the transverse separation).

2.2 The Pocklington kernel

The resolution, due to Pocklington [1], is to combine the scalar and vector potential contributions *before* evaluating the derivative, so that the near-field blow-ups cancel. Writing $u = z - z'$, ρ the transverse separation (equal to a for the self term, to the wire spacing for a mutual term), and $R = \sqrt{\rho^2 + u^2}$, one finds by direct differentiation

$$G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2} = \frac{e^{-jkR}}{4\pi}\left[\frac{\rho^2}{R^3} + j\frac{2u^2 - \rho^2}{kR^4} + \frac{2u^2 - \rho^2}{k^2 R^5}\right],$$

and hence

$$-j\omega\mu_0\left[G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2}\right] = \frac{j\omega\mu_0}{4\pi}\,K(z, z'),$$

where the **Pocklington kernel** is

$$\boxed{K = e^{-jkR}\left[\frac{u^2}{R^3} - \frac{2j\,u^2}{kR^4} - \frac{2\,u^2}{k^2R^5} + \frac{j\,\rho^2}{kR^4} + \frac{\rho^2}{k^2R^5} - \frac{1}{R}\right].} \tag{3}$$

Two structural facts about (3) matter for the numerics.

No separate endpoint term. The scalar-potential (charge) contribution is already *inside* the kernel integral. This is not an accident: it is the fundamental theorem of calculus. The charge density is $\rho_c(z') = (j/\omega)\,dI/dz'$ (continuity equation), and its potential contribution to E_z^s is $-\partial\Phi_c/\partial z$, which integrates by parts to a boundary term $g(z_1) - g(z_2) = -\int (dg/dz')dz'$ that is absorbed into the G term. Adding a *separate* endpoint/charge term therefore double-counts the scalar potential and corrupts the reactance. This is the first pitfall in constructing a thin-wire MoM code.

A sharp, narrow self-peak. As $R \to \rho$ (observation point on the surface directly above a source point, $u \to 0$), the kernel is dominated by the ρ^2/R^3 and $\rho^2/(k^2R^5)$ pieces, which peak with a width of order $\rho = a$. The self-term ($m = n$) matrix element therefore integrates a feature of width a — for $a = 3$ mm and a segment of length $\Delta\ell \sim 1$–5 mm the peak is comparable to or broader than the segment. This is the second, and numerically more serious, pitfall: a single coarse quadrature under-integrates the diagonal peak and collapses the impedance toward zero. §4 treats the quadrature this demands.

2.3 The EFIE as an operator

For the functional-analytic reader it is useful to recast (2) as an operator equation. Let $\mathcal{K}$ be the integral operator

$$(\mathcal{K}I)(z) = \frac{j\omega\mu_0}{4\pi}\int_{-L/2}^{L/2} K(z, z')\,I(z')\,dz',$$

acting on the space of (square-integrable) currents on the wire. The EFIE is

$$\mathcal{K}\,I = E_z^i. \tag{4}$$

$\mathcal{K}$ is a compact (Hilbert-Schmidt, modulo the weak $1/R$ singularity which is integrable) operator on L^2, so (4) is a regularised first-kind equation; the MoM replaces it by a finite-rank approximation. The choice of basis (the range of the finite-rank operator) and of testing (collocation vs. Galerkin) determines the discretisation error and, as we argue in §3.4, the *validity* of the kernel pairing.

3. Method-of-moments discretisation

3.1 Basis and expansion

We expand the current in **piecewise-constant** (PW-const) basis functions. The wire $[-L/2, L/2]$ is split into N segments of length $\Delta\ell = L/N$ with nodes $z_n = -L/2 + n\,\Delta\ell$, and

$$I(z) = \sum_{n=0}^{N-1} I_n\,\chi_n(z), \qquad \chi_n(z) = \begin{cases} 1, & z \in [z_n, z_{n+1}) \\ 0, & \text{otherwise.} \end{cases}$$

The coefficients I_n are the unknowns. The PW-const basis is the simplest choice and, with $\Delta\ell \ll \lambda$, is spectrally adequate for the half-wave regime that matters here. (The piecewise-*sinusoidal* basis, $f_n(z) = \sin(k(z - z_n))$, is a natural refinement — it enforces a smoother, approximately-continuous current and is more accurate in the short-dipole limit — but it changes the testing requirement, as discussed in §3.4.)

3.2 Collocation (point matching)

We enforce the EFIE **at a point**: at the centre z_m of each segment m, with the test point offset one radius a from the axis (on the wire surface). Substituting the PW-const expansion,

$$\sum_{n=0}^{N-1} I_n \underbrace{\frac{j\omega\mu_0}{4\pi}\int_{z_n}^{z_{n+1}} K(z_m, z')\,dz'}_{A_{mn}} = E_z^i(z_m),$$

gives the linear system

$$\mathbf{A}\,\mathbf{I} = \mathbf{b}, \qquad A_{mn} = \frac{j\omega\mu_0}{4\pi}\int_{z_n}^{z_{n+1}} K(z_m, z')\,dz', \tag{5}$$

where the right-hand side $\mathbf{b}$ is the incident field sampled at the test points. For a delta-gap feed of voltage V_0 at the centre, $E_z^i = V_0\,\delta(z)$, so $b_{\text{feed}} = V_0$ (dispersed as $V_0/\Delta\ell$ over the feed segment in the point-matching form) and $b_m = 0$ elsewhere.

3.3 The impedance matrix and the input impedance

The matrix $\mathbf{A}$ in (5) is the MoM impedance matrix. It is dense, complex, and non-symmetric in general (the kernel $K(z, z')$ is not symmetric in its arguments once the observation point is offset from the source axis). Its construction costs $O(N^2)$ kernel integrals, each a one-dimensional quadrature; the solve is $O(N^3)$ (or $O(N^2)$ per frequency for a fixed geometry with a factorised matrix).

The **input impedance** follows from the delta-gap model. With the PEC condition $E^s = -E^i$, the solved current on the feed segment I_{feed} relates to the applied voltage by

$$\boxed{Z_{\text{in}} = -\frac{V_0}{I_{\text{feed}}}} \tag{6}$$

The minus sign is a consequence of $E^s = -E^i$ and is a common source of sign errors: using $+V_0$ flips the sign of the radiation resistance, and a *negative* R_{rad} is the obvious red flag that the excitation sign is wrong.

For a **multi-wire** structure (a driven element plus open-circuited parasitics) the EFIE is enforced on *all* segments. The source is non-zero only at the driven element's feed segment; the parasitic segments have $E^i = 0$ (PEC, open circuit), so their currents are *induced* by the driven element. The feedpoint impedance of the driven element is Z_{in} from the full-system solve (6). For a true two-port (both elements fed) the open-circuit Z-parameters follow from the unit-current method: drive port p with $I_{\text{feed},p} = 1$ (replacing that row of $\mathbf{A}$ by the linear constraint $I_{\text{feed},p} = 1$), and read off $Z_{qp} = V_q / I_{\text{feed},p}$. A loaded two-port then reduces to $Z'_{11} = Z_{11} - Z_{12}Z_{21}/(Z_{22} + Z_L)$.

3.4 Why collocation, not Galerkin, with this kernel

This is the subtle point that the earlier (buggy) version of the solver got wrong, and it is worth stating precisely because it is a general feature of integral-equation discretisations.

The Pocklington kernel (3) is derived so that the EFIE (2) holds **pointwise** in the observation coordinate z. It is therefore the correct kernel for *collocation* (test at a point). If instead one multiplies (2) by a test function $f_m(z)$ and integrates over z (Galerkin), the term $(1/k^2)\,\partial^2 G/\partial z^2$ must be integrated by parts:

$$\int f_m(z)\,\frac{\partial^2 G}{\partial z^2}\,dz = \Big[f_m \frac{\partial G}{\partial z} - f'_m\,G\Big]_{\text{endpoints}} + \int f''_m(z)\,G\,dz.$$

Two things happen: (i) the second z-derivative on G is replaced by $f''_m(z)G$ (a derivative on the *test function*, not on the Green's function), and (ii) **endpoint terms** $[f_m\,\partial G/\partial z - f'_m G]$ appear. Feeding the un-integrated Pocklington kernel directly into a double (Galerkin) integral — $A_{mn} = \iint f_m K f_n\,dz\,dz'$ — skips *both* steps. It is not a discretisation of (2); it is a different, wrong operator.

This is exactly what the first version of `antenna_mom.py` did: a piecewise-sinusoidal basis with a Galerkin (double-integral) assembly of the un-integrated kernel. For a half-wave dipole it returned $Z \approx 0 - j1500\ \Omega$ instead of $\approx 73 + j42\ \Omega$, and the error did not converge with N (it is a formulation error, not a discretisation error, so mesh refinement cannot fix it). Replacing the Galerkin assembly by collocation with the Pocklington kernel (5) — the change made in the corrected solver — restores agreement with the classical values to $< 1\%$ in the half-wave regime (§4).

Lesson. The kernel and the testing procedure are a *pair*. The Pocklington kernel (3) is the collocation kernel. A Galerkin discretisation requires the integrated-by-parts kernel (with $f''_m G$ and endpoint terms), which is a different object. Mixing them is a category error that no amount of quadrature refinement can repair.

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4. Numerical analysis and validation

4.1 The quadrature: two-level Gauss–Legendre

The kernel (3) is smooth and bounded for $u \neq 0$ but peaks sharply at $u = 0$ on the **self** segment, with a width of order a. The off-diagonal terms ($m \neq n$) are smooth over the source segment. The correct response is a *two-level* quadrature:

- **Diagonal (self, $m = n$):** fine Gauss–Legendre, $n_q^{\text{self}} \approx 300$–$400$ nodes, to resolve the width-a peak.
- **Off-diagonal ($m \neq n$):** coarse Gauss–Legendre, $n_q^{\text{off}} \approx 24$ nodes, since the integrand is smooth.

A single coarse quadrature (say $n_q = 24$ everywhere) under-integrates the diagonal peak. In the buggy version this collapsed Z_{in} toward zero; with the corrected two-level scheme the half-wave anchors are reproduced to $< 1\%$ (§4.2). The cost of the fine diagonal is modest: only N of the N^2 elements use n_q^{self}.

4.2 Validation against the classical limits

The solver is validated against a battery of classical results before any design number is trusted. All results below are produced by `validate_fixed.py` at $f = 144$ MHz ($\lambda \approx 2.08$ m) unless stated.

(a) Half-wave dipole anchors. A centre-fed dipole of length 0.47λ and radius 3 mm should give $Z_{\text{in}} \approx 70 - j10\ \Omega$; one of 0.50λ and radius 1 mm should give $\approx 81 + j27\ \Omega$. The solver returns

Geometry	Z_{in} **(solver)**	**Reference**	**SWR @ 50** Ω
0.47λ, $a = 3$ mm, $N = 201$	$70.27 - j10.20\ \Omega$	$\approx 70 - j10$	1.46
0.50λ, $a = 1$ mm, $N = 201$	$81.39 + j27.52\ \Omega$	$\approx 81 + j27$	1.90

Agreement is to well under 1 % in both real and imaginary parts.

(b) Convergence in N. For the 0.47λ dipole:

N	Z_{in}	$\lvert\Delta Z\rvert$ **vs** N=401
101	$68.28 - j26.67$	2.34×10^{1}
201	$70.27 - j10.20$	6.80

301	$71.34 - j4.00$	5.32×10^{-1}
401	$71.56 - j3.52$	(ref)

The sequence converges monotonically; by $N = 301$ the change on further refinement is $< 0.6\ \Omega$. This is the normal PW-const convergence behaviour in the half-wave regime.

(c) Resonant length. Scanning L for the zero of the reactance gives $L/\lambda \approx 0.482$, against the classical 0.473 — a 2% length error, consistent with the finite radius ($a = 3$ mm shortens the resonant length).

(d) Power balance. For a unit feed current the input power $P_{\text{in}} = \frac{1}{2}\,\text{Re}(Z_{\text{in}})$ must equal the radiated power P_{rad} computed from the far-field Poynting integral (up to conductor loss). The solver gives $P_{\text{in}} = 35.1365$ W, $P_{\text{rad}} = 35.1372$ W, ratio 1.00002. This near-exact balance is a strong self-consistency check: it confirms that the impedance extraction (6) and the far-field normalisation are mutually consistent. It holds at all lengths tested, including the electrically-long dipoles of §4.3.

(e) Directivity and pattern. The 0.47λ dipole gives $D = 1.634$ (2.13 dBi) against the classical 2.15 dBi, an E-plane 3-dB beamwidth of 79° against $\approx 78^\circ$, and a radiating efficiency of 99.88% (copper surface resistance). All within a few percent of the textbook values.

4.3 Where the piecewise-constant basis degrades

Two limits expose the limitations of the PW-const basis. These are *basis* limitations, not kernel bugs — the power balance remains 1.0000 in both cases, so the solver is self-consistent; the issue is that the PW-const current is a poor representation of the true current in these limits.

(a) The short-dipole limit ($L \ll \lambda$). The radiation resistance of a short centre-fed dipole with (approximately) sinusoidal, feed-normalised current is $R_{\text{rad}} = 20\pi^2(L/\lambda)^2$ (the factor $80\pi^2$ belongs to the *uniform*-current Hertzian idealisation and must not be used as the comparison). The solver under-predicts R_{rad}, and the error shrinks as the dipole lengthens:

L/λ	R_{rad} **(solver)**	$20\pi^2(L/\lambda)^2$	**ratio**
0.02	0.0000	0.0790	0.00
0.05	0.1246	0.4935	0.25
0.10	1.4286	1.9739	0.72

The cause is visible in the current profile: for $L = 0.02\lambda$ the solved current is a narrow spike at the feed (peaking at 0.89 of its max and falling to ~ 0 within a segment or two) rather than a triangle spanning the wire. The PW-const staircase does not spread the

current over the short wire, so it radiates less than the analytical triangle. A piecewise-sinusoidal basis (which enforces a smoother, end-vanishing current) removes most of this error; we do not need it for the half-wave design of §5.

(b) The electrically-long limit ($L \gtrsim 0.7\lambda$). For a 0.75λ dipole the solved *current distribution* converges and has the correct "W" shape (a dip at the feed and two peaks, the standing-wave pattern of a 0.375λ arm), but the radiation resistance does **not** converge in N:

N	Z_{in}
101	$603 + j571\ \Omega$
201	$624 + j564\ \Omega$
401	$837 + j540\ \Omega$
601	$1049 + j391\ \Omega$

The real part climbs without settling as the mesh is refined. The power balance is still 1.0000 (the solver is self-consistent), so this is not a bookkeeping error; it is a genuine failure of the PW-const collocation discretisation to converge for electrically-long, weakly-ill-conditioned currents. This matters for §5 because, on the 70 cm band, the 2 m driven element is 1.36λ long and sits squarely in this regime. We treat the 70 cm feedpoint numbers as *indicative* (correct to order of magnitude and in sign) rather than converged, and we say so explicitly where they are used.

The pulse-basis convergence caveat. This non-convergence is not a bug in the implementation; it is a known property of the piecewise-*constant* (pulse) basis with point-matching on the thin-wire Pocklington equation. The discretisation is only *conditionally* convergent: it converges for $\Delta\ell \gtrsim a$ (segment length comparable to or larger than the wire radius) but not in the limit $\Delta\ell \to 0$ once $\Delta\ell < a$, because the pulse basis cannot represent the smooth, end-vanishing current — and, in particular, the charge $\partial I/\partial z'$ — that a long wire carries. The standard remedy is a **piecewise-linear (rooftop/triangle) basis**, which is C^0-continuous across segment boundaries and represents the current derivative (charge) naturally; with a rooftop basis the collocation discretisation converges as $\Delta\ell \to 0$. We retain the pulse basis here because it is adequate in the half-wave regime ($0.2 \lesssim L/\lambda \lesssim 0.6$) where the design elements live on their own bands, and the degradation is confined to the electrically-long 70 cm feedpoint, which we report as indicative.

4.4 Summary of the validated regime

The corrected solver is reliable — to well under a few percent — for centre-fed dipoles and dipole-like elements in the range $0.2 \lesssim L/\lambda \lesssim 0.6$, i.e. the half-wave regime. This is exactly the regime of the dual-band design in §5 *on each element's own band*. The 70 cm feedpoint, which depends on the 2 m element at 1.36λ, is outside the validated regime and is reported as indicative.

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5. Applied design: a dual-band VHF/UHF open-sleeve dipole

5.1 The design problem

We design a single-structure antenna for the amateur 2 m band (144–148 MHz) and 70 cm band (430–440 MHz). A single straight dipole cannot be resonant at both bands — the frequencies are a factor of three apart, so a half-wave dipole would need to be simultaneously ≈ 1 m and ≈ 0.35 m long. The classic resolution is the **open-sleeve dipole**: a centre-fed 2 m driven dipole (resonant near 145 MHz) with 70 cm sleeves (resonant near 435 MHz) placed *very close* to the driven element, typically as a symmetric pair on both sides. The sleeves are open-circuited; their current is induced by the driven element.

The essential physics is the **coupling regime**. A sleeve placed in the reactive near-field of the driven element — a spacing of order 10–25 mm, i.e. 0.015–0.035λ at UHF — couples *strongly* (the mutual impedance $|Z_{12}|$ is tens of ohms) and acts as an impedance transformer. A second element placed at a far-field (Yagi-like) spacing of ~ 0.2– 0.5λ couples *weakly* ($|Z_{12}|$ of a few ohms) and shapes the pattern without transforming the feedpoint impedance. The open-sleeve dipole is the first case; confusing the two is the central error we examine in §5.4. On the 2 m band the driven element radiates and the (electrically short) sleeves are a weak perturbation; on the 70 cm band the sleeves are near-resonant and strongly coupled, and they match the feedpoint. The canonical parametric study is King and Wong [4], who measured VSWR as a function of sleeve length and *sleeve-to-dipole spacing* and found the dual-band match to require close coupling (spacing of order 0.02–0.04λ on the sleeve band) — the reactive near-field, not the Yagi-like spacing.

5.2 Geometry

The driven element is a 2 m dipole; the 70 cm band is served by a **symmetric pair** of sleeves close-coupled to it:

- **2 m driven element:** $L_2 = 936$ mm, radius $a_2 = 3$ mm, at $x = 0$, centre-fed. ($L_2 = 0.452\lambda$ at 145 MHz.)
- **70 cm sleeves (pair):** $L_7 = 316$ mm each, radius $a_7 = 3$ mm, at $x = \pm 25$ mm. ($L_7 = 0.458\lambda$ at 435 MHz; spacing 25 mm $= 0.036\lambda$.)
- Mesh: $N_2 = 201$ segments on the driven element, $N_7 = 61$ on each sleeve; two-level quadrature $n_q^{\text{self}} = 400$, $n_q^{\text{off}} = 24$.

The full three-wire system ($N_2 + 2N_7 = 323$ unknowns) is solved at each frequency; the feedpoint is on the driven element. The sleeve length and spacing were optimised (sweeping $L_7 \in \{300, 316, 332\}$ mm and $x_7 \in \{20, 25, 30\}$ mm) to minimise the 70 cm SWR at 435 MHz; the optimum is $L_7 = 316$ mm, $x_7 = 25$ mm.

5.3 Feedpoint impedance and SWR

The solver (`dualband_design_v2.py`) sweeps both bands. The close-coupled geometry gives:

2 m band (144–148 MHz).

f **(MHz)**	Re Z_{in}	Im Z_{in}	**SWR @ 50** Ω
144	56.8	-45.5	2.31
145	58.6	-40.6	2.11

146	60.4	-35.7	1.94
147	62.3	-30.8	1.80
148	64.2	-25.9	1.67

The 2 m band is moderately matched (SWR 1.7–2.3) and *improves* with frequency: the driven element is 0.452λ (slightly short of the 0.473λ resonance), so it is capacitive, and the reactance falls toward zero as the band centre is approached from below. The close-coupled sleeves are electrically short on this band (0.15λ) and perturb the match only slightly. This is the well-validated regime (§4.4), so these numbers are reliable.

70 cm band (430–440 MHz).

f **(MHz)**	Re $Z_{\mathbf{in}}$	Im $Z_{\mathbf{in}}$	**SWR @ 50** Ω
430	59.2	-25.7	1.64
432	59.9	-22.7	1.57
434	60.6	-19.7	1.50
436	61.3	-16.8	1.44
438	62.0	-13.9	1.39
440	62.6	-11.1	1.35

The 70 cm band is now **well matched** (SWR 1.35–1.64). The large capacitive reactance of the 1.36λ driven element has been almost entirely cancelled by the close-coupled sleeves, leaving a feedpoint near $61 - j18\ \Omega$ at the band centre. This is the result the open-sleeve concept is supposed to give — and it is what distinguishes the close-coupled design from the far-field (Yagi-like) spacing examined in §5.4.

5.4 How the close-coupled sleeve matches the 70 cm band

The mechanism is the two-port (here three-port) impedance transformation. With the driven element as port 1 and the two sleeves as ports 2 and 3, the open-circuit

Z-parameters at 435 MHz are

Parameter	**Value**	**Meaning**
Z_{11}	$63.0 - j130.2\ \Omega$	driven-element self-impedance (sleeves open)
Z_{12}	$22.6 - j68.1\ \Omega$, $\lvert Z_{12}\rvert = 71.8\ \Omega$	mutual, driven–sleeve +
Z_{13}	$17.9 - j61.7\ \Omega$, $\lvert Z_{13}\rvert = 64.2\ \Omega$	mutual, driven–sleeve −
Z_{22}	$24.3 - j42.6\ \Omega$	sleeve self-impedance (near-resonant)
Z_{23}	$22.2 - j22.8\ \Omega$	mutual, between the two sleeves

($|Z_{12}|$ and $|Z_{13}|$ are equal by symmetry; the small difference is a test-point-offset artefact of the discretisation.)

The loaded feedpoint, with both sleeves open, is

$$Z'_{\text{in}} = Z_{11} - Z_{1v}\, Z_{vv}^{-1}\, Z_{v1} = 60.9 - j18.2\ \Omega, \qquad \text{SWR} = 1.47,$$

where $Z_{1v} = [Z_{12}\ Z_{13}]$ and Z_{vv} is the 2×2 sleeve sub-matrix. The physics is immediate: the driven element alone is $63 - j130\ \Omega$ (the 1.36λ capacitive reactance), and the close-coupled, near-resonant sleeves contribute a correction of roughly $+3 - j128\ \Omega$ — a large *negative* reactance that almost exactly cancels the driven element's. The near-resonant sleeve presents a low- impedance path; because it is in the reactive near-field, the mutual impedance is large ($|Z_{12}| \approx 70\ \Omega$, not a few ohms), so the cancellation is strong.

The coupling-regime contrast. The same driven element with a *single* 70 cm element at a far-field (Yagi-like) spacing of 307 mm (0.445λ) gives $|Z_{12}| \approx 3\ \Omega$ and leaves the feedpoint at $67.8 - j137.6\ \Omega$ (SWR 7.55) — the weakly-coupled element cannot transform the impedance. Sweeping the spacing of a *single* offset sleeve shows the transition: at 307 mm the SWR is 7.55; at 25 mm it falls to 2.0; but a single offset sleeve also pulls the real part down ($24\ \Omega$ at 15 mm), so it never reaches SWR < 1.5. It is the **symmetric pair** in the reactive near-field that both cancels the reactance *and* preserves the real part near $50\ \Omega$:

Sleeve placement (70 cm)	$\lvert Z_{12}\rvert$	$Z'_{\textbf{in}}(435\ \textbf{MHz})$	**SWR**

single, $+307$ mm (0.445λ)	$3.0\ \Omega$	$67.8 - j137.6$	7.55
single, $+25$ mm (0.036λ)	$73.6\ \Omega$	$54.4 - j37.1$	2.02
pair, ± 25 mm (0.036λ)	$71.8\ \Omega$	$60.9 - j18.2$	**1.47**
pair, ± 20 mm (0.029λ)	$79.4\ \Omega$	$41.2 - j16.4$	1.50

The optimum is ± 25 mm (SWR 1.47); tighter spacing (± 10 mm) over- couples and collapses the real part (SWR 5.5). This is the open-sleeve mechanism, and it is entirely a consequence of the reactive near-field coupling — it would be impossible at the far-field spacing.

5.5 Radiation pattern, directivity, and gain

Plane definitions. For a z-directed dipole the **E-plane** is an elevation cut — any plane containing the wire axis; we take $\phi = 0$ (the x-z plane), θ swept $0 \to 360°$. The **H-plane** is the *azimuthal* cut at broadside — $\theta = 90°$ (the x-y plane), ϕ swept $0 \to 360°$. It is a common error to take a second elevation cut ($\phi = 90°$) and call it the H-plane; that cut still shows the end-fire nulls and is not the H-plane. The H-plane of a vertical dipole must be (nearly) **omnidirectional** — a circle — and never has end-fire nulls.

The far field is the coherent sum of Hertzian elements (§3.3, Appendix A). For the close-coupled geometry the pattern metrics are:

Quantity	**145 MHz (2 m)**	**435 MHz (70 cm)**
Directivity D	1.98 dBi	3.23 dBi

Gain G	1.98 dBi	3.23 dBi
Radiating efficiency	99.9%	100%
E-plane 3-dB beamwidth	84°	66°
H-plane 3-dB beamwidth	360° (omni)	360° (omni)
Front-to-back ratio	0.0 dB	1.7 dB

On the 2 m band the pattern is that of a half-wave dipole: E-plane directivity 1.98 dBi (classical 2.15 dBi), an E-plane beamwidth of 84° (classical $\approx 78^\circ$), and an omnidirectional H-plane.

On the 70 cm band the E-plane is that of the electrically-long driven element (1.36λ) loaded by the sleeves: a higher directivity (3.23 dBi) and a narrower E-plane beam (66°). The H-plane remains essentially omnidirectional (360°) on both bands, as a z-directed structure requires; the small front-to-back ratio (1.7 dB at 70 cm) is the only azimuthal asymmetry, set by the sleeve pair. Radiating efficiency is $\approx 100\%$ on both bands (copper loss in a 3 mm wire at VHF/UHF is negligible), so gain equals directivity.

The polar patterns are in `patterns_dualband.png`: the E-plane is a figure-eight (lobes broadside, nulls end-fire) and the H-plane is a (nearly) perfect circle on both bands.

5.6 Design assessment

The close-coupled open-sleeve dipole is a genuinely dual-band antenna: well-matched on the 2 m band (SWR 1.7–2.3) *and* on the 70 cm band (SWR 1.35–1.64), with near-omnidirectional H-plane patterns and $\approx 100\%$ radiating efficiency on both. The MoM analysis quantified the design before any metal was bent and, more importantly, *localised the physics*: the 70 cm match is produced by reactive near-field coupling ($|Z_{12}| \approx 70\ \Omega$) between the driven element and the sleeve pair, and it would be impossible at a far-field (Yagi-like) spacing where $|Z_{12}|$ is a few ohms. The lesson is that a dual-band "parasitic" is not a generic second element — its spacing relative to λ determines whether it transforms the feedpoint impedance or merely shapes the pattern.

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6. Discussion

The exercise illustrates three points that are general to integral-equation antenna analysis.

Formulation errors do not converge. The Galerkin-with-unintegrated-kernel bug of §3.4 produced a Z_{in} that was wrong by a factor of ~ 20 and *did not improve with mesh refinement*, because it discretised the wrong operator. This is a caution for any MoM code: a result that does not converge with N is a signal of a formulation (or sign) error, not a discretisation error, and no quadrature refinement will fix it. The correct diagnostic is to check the *classical limits* (half-wave impedance, short-dipole R_{rad}, power balance) before trusting any design number.

Basis choice sets the validated regime. The PW-const basis is accurate in the half-wave regime ($0.2 \lesssim L/\lambda \lesssim 0.6$) and degrades in the short- and long-dipole limits (§4.3). The power balance staying at 1.0000 even where the impedance is wrong shows that self-consistency (Poynting balance) is necessary but not sufficient for accuracy — it checks the *normalisation*, not the *current distribution*.

A single number can hide the physics. A 70 cm SWR of 7.55 (the far-field-spaced element of §5.4) looks like a "bad design," but the three-port Z-parameters localise the cause: it is the driven element at 1.36λ, and the element is too *weakly* coupled ($|Z_{12}|$ a few ohms) to transform the impedance. Move the same element into the reactive near-field and $|Z_{12}|$ jumps to $\approx 70\,\Omega$ and the SWR drops to 1.47. The MoM result is not just a pass/fail SWR; it is a diagnostic of *where* the match or mismatch lives — and of the coupling regime that produces it.

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7. Conclusion

We derived the thin-wire Pocklington EFIE, discretised it by MoM with piecewise-constant basis and collocation testing, and analysed the discretisation's singular structure, quadrature requirements, and convergence. The corrected solver reproduces the classical half-wave dipole impedance, resonant length, directivity, and power balance to well under a few percent, and its validated regime ($0.2 \lesssim L/\lambda \lesssim 0.6$) covers each element of a dual-band VHF/UHF open-sleeve dipole on its own band. Applied to a 144–440 MHz design, the solver predicts a close-coupled open-sleeve dipole that is well-matched on *both* bands — 2 m SWR 1.7–2.3 and 70 cm SWR 1.35–1.64 — and, through the three-port Z-parameters, shows that the 70 cm match is produced by reactive near-field coupling ($|Z_{12}| \approx 70\ \Omega$) between the driven element and a symmetric sleeve pair, and would be impossible at a far-field (Yagi-like) spacing where $|Z_{12}|$ is a few ohms. The method is a reliable measuring instrument for wire-antenna design in the half-wave regime, and its limitations (short- and long-dipole basis degradation, and the pulse-basis convergence caveat of §4.3) are clearly identifiable.

References

[1] H. Pocklington, "On the Calculation of Electrostatic and Electromagnetic Problems by Means of a System of Linear Equations," *Proc. Phys. Soc. London*, ser. A, vol. 36, pp. 537–552, 1924. (The original method-of-moments formulation for wire antennas; the kernel (3) is the Pocklington kernel.)

[2] R. F. Harrington, *Field Computation by Moment Methods*. New York: IEEE Press / W. H. Freeman, 1968. (The standard treatment of the MoM: basis and testing functions, the EFIE, and the thin-wire kernel.)

[3] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ: Wiley, 2016. (Classical dipole impedances, the short-dipole $R_{\text{rad}} = 20\pi^2(L/\lambda)^2$, and the half-wave dipole reference values used in §4.)

[4] H. King and J. Wong, "An experimental study of a balun-fed open-sleeve dipole in front of a metallic reflector," *IEEE Trans. Antennas Propagat.*, vol. 20, p. 201, 1972. (The classical parametric study of the open-sleeve dipole: VSWR as a function of sleeve length and sleeve-to-dipole spacing — the basis for the close-coupled (reactive near-field) spacing used in §5.)

[Note on citation verification: references [1]–[4] are standard, well-established works. For [4] the journal, volume, starting page, and year are from the ADS record (bibcode 1972ITAP..20..201K); the full page range should be confirmed against the IEEE Xplore record. The other references have specific page numbers intentionally omitted; the results attributed to them (the Pocklington kernel, the MoM discretisation, the classical

dipole impedances) are the canonical content of each text. Verify page-level detail against the physical editions before formal publication.]

Appendix A: The solver

`antenna_mom.py` (NumPy only) implements:

- `Wire(x0, y0, z1, z2, a, N)` — a straight PEC wire.
- `build_Z(wires, f, nq_off, nq_self)` — the MoM impedance matrix (5), two-level Gauss–Legendre quadrature, Pocklington kernel (3), no endpoint term.
- `solve(wires, f, feed_seg, V0)` — (Z_{in}, I) from the delta-gap model (6), $Z_{\text{in}} = -V_0/I_{\text{feed}}$.
- `z_parameters(wires, f, feed_segs)` — open-circuit n-port Z-parameters (unit-current method).
- `far_field`, `radiated_power`, `directivity`, `efficiency`, `swr`, `s11`, `pattern_metrics` — far-field and pattern quantities. `pattern_metrics` returns the E-plane (an elevation cut, $\phi = 0$, θ swept) and the H-plane (the *azimuthal* cut at broadside, $\theta = 90^\circ$, ϕ swept) as separate 1-D arrays, so the two planes are not confused (§5.5).

The validation battery `validate_fixed.py` runs the checks of §4; `dualband_design_v2.py` runs the close-coupled open-sleeve design of §5 (sleeve-length/spacing optimisation, both band sweeps, the three-port Z-parameters of §5.4, and the pattern metrics of §5.5) and writes `dualband_v2_results.json`. `make_patterns.py` renders `patterns_dualband.png` — the E-plane (elevation) and H-plane (azimuthal) polar cuts at 145 and 435 MHz, with the sleeve directions marked.